Smart Horizon 2026:
48-Hour International Hackathon

Level-1

Organized by: Dept. of Artificial Intelligence and Machine Learning & Dept. of Computer Science and Engineering

Event Date	Venue	Host Institute
September 3rd, 4th & 5th, 2026	New Horizon Knowledge Park, Bengaluru	New Horizon College of Engineering

Title: LUNABOT - Smart Autonomous Moon Rover

Team Members
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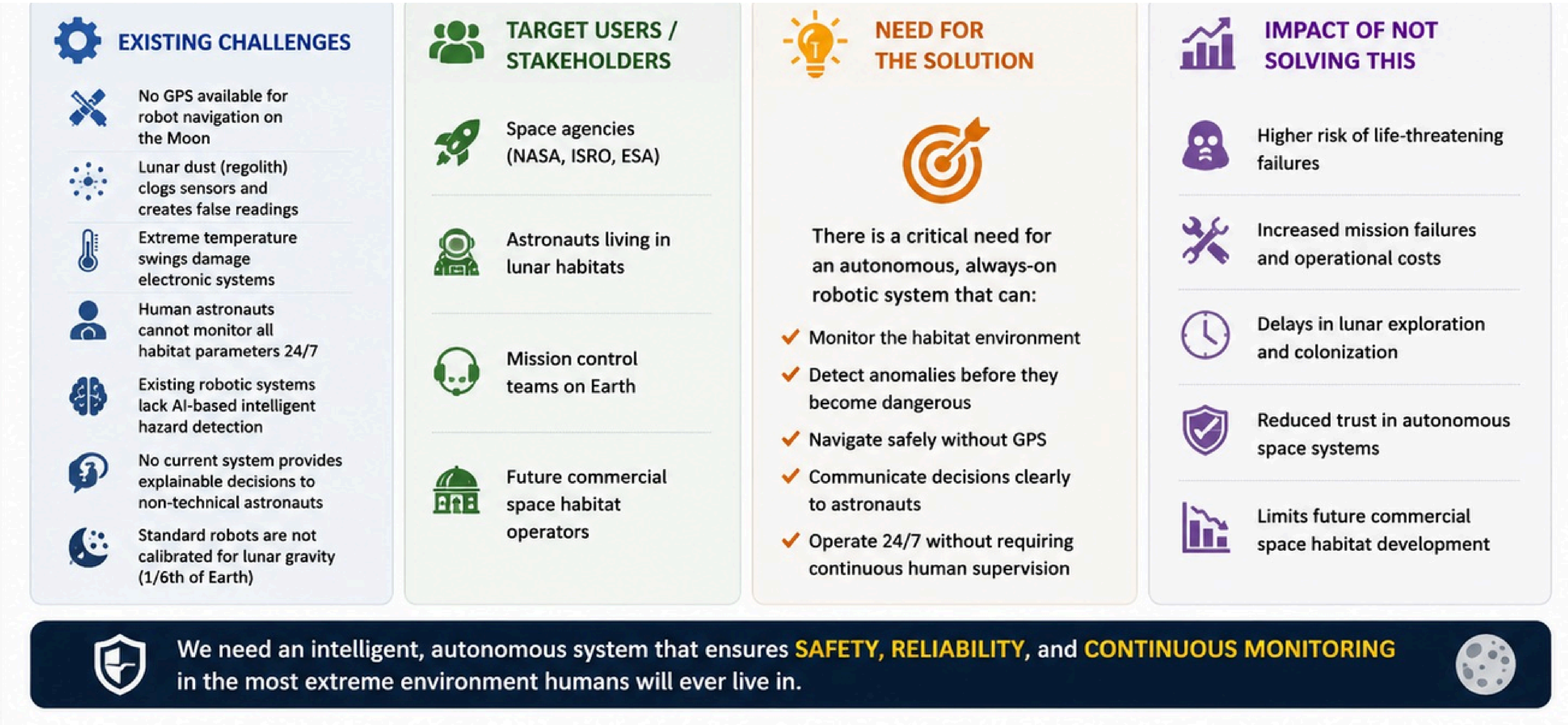
Team Details
Team ID : SHIH26-TID-387
Team Name : TECHTONICS
Team Lead: AKSHAYA SRELEKA P S
College Name: CHENNAI INSTITUTE OF TECHNOLOGY
Problem Statement ID: SH-DST-04

PROBLEM UNDERSTANDING

PROBLEM DESCRIPTION

Lunar habitats are among the most dangerous environments ever designed for human survival. Astronauts face constant life-threatening risks-oxygen leaks, CO, buildup, pressure drops-in an environmentwith no atmosphere, no GPS, and temperatures dropping to -173°C. A single undetected anomaly can be fatal.



LIMITATIONS IN EXISTING SOLUTIONS

- Depend heavily on manual monitoring and human intervention
- Use fixed-threshold alarms, causing false alerts or missed hazards
- Cannot operate effectively in GPS-denied lunar environments
- Poor adaptation to lunar dust, low gravity, and extreme temperatures
- Lack integrated AI-based hazard detection, autonomous navigation, and explainable decision-making

NEED FOR THE SOLUTION

- Continuous monitoring of critical habitat conditions for astronaut safety
- Early AI-based detection of hidden or complex hazards
- Autonomous operation in GPS-denied and extreme lunar environments
- Reduced human workload through intelligent monitoring and navigation

PROPOSED SOLUTION & INNOVATION

COMPARISON: EXISTING SOLUTION VS LUNABOT

FEATURE	EXISTING SYSTEMS	LUNABOT (OUR SOLUTION)
Dust handling	✗ No filtering	✓ Adaptive LiDAR filter
Anomaly detection	✗ Simple thresholds	✓ Isolation Forest AI
Night operation	✗ System shutdown	✓ 3-mode survival system
Decision transparency	✗ No explanation	✓ Full XAI in plain English
Position correction	✗ Drifts over time	✓ ArUco EKF correction
Human interaction	✗ Dashboard only	✓ Natural language queries

SOLUTION OVERVIEW

Autonomous Lunar Navigation System

ROS2-based robot capable of operating in GPS-denied lunar habitats, performing mapping, localisation, and safe navigation using multi-sensor fusion

Robust Perception & Mapping Pipeline

LiDAR, camera, and environmental sensors collect data → dust filtering ensures clean inputs → SLAM builds accurate maps and position estimates in real time

AI-Driven Hazard Detection & Safety Mapping

ML model detects environmental anomalies → hazard zones are dynamically identified and permanently marked in the navigation map

Intelligent & Safe Navigation

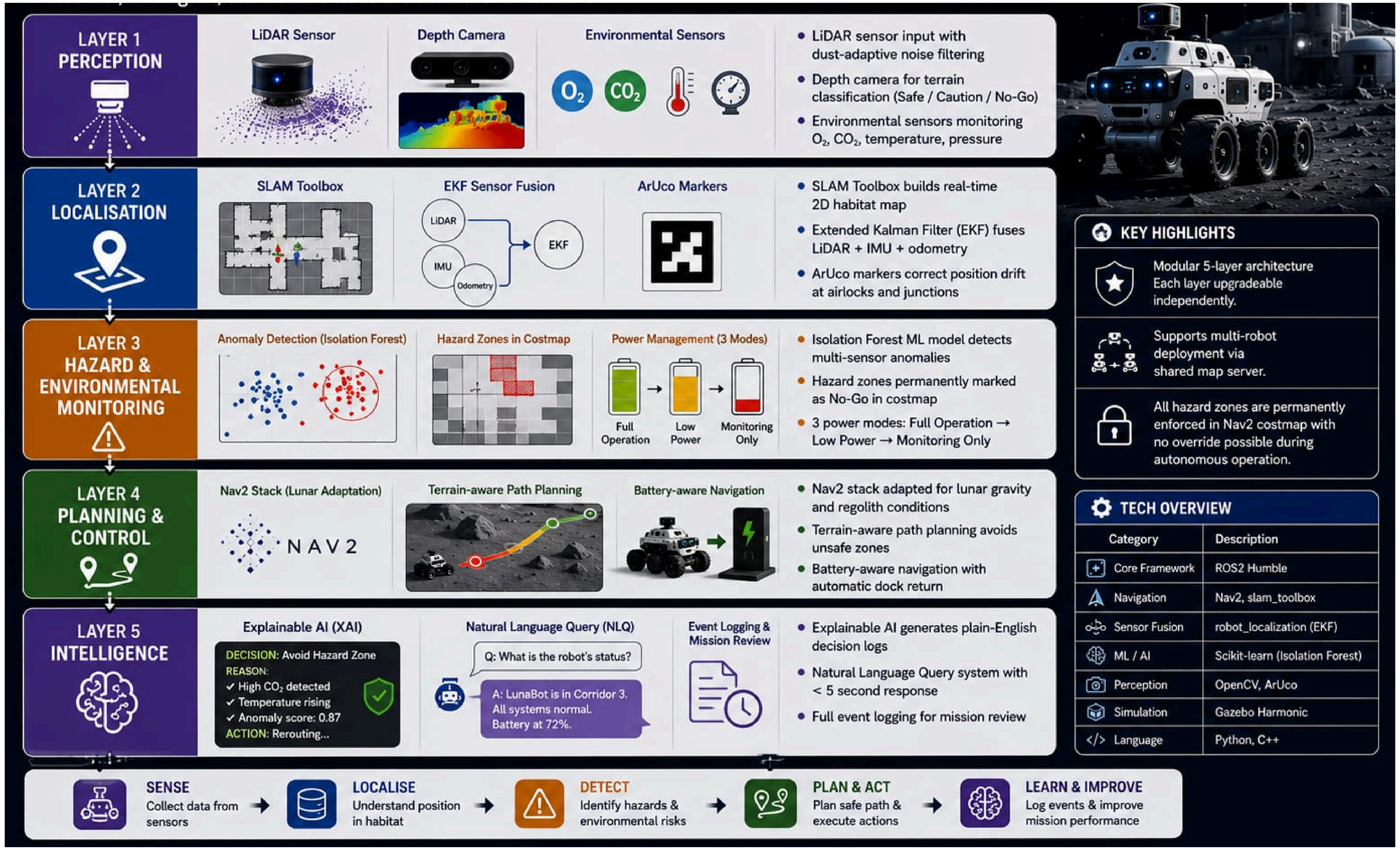
Navigation system avoids unsafe terrain & hazard zones while performing continuous autonomous patrol with adaptive decision-making

Explainable & Simulation-Validated System

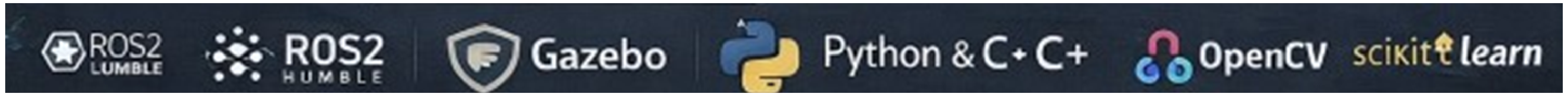
Every action is explained via Explainable AI→ fully validated in Gazebo lunar simulation with realistic gravity, regolith, and temperature conditions

TECHNICAL ARCHITECTURE

SYSTEM ARCHITECTURE



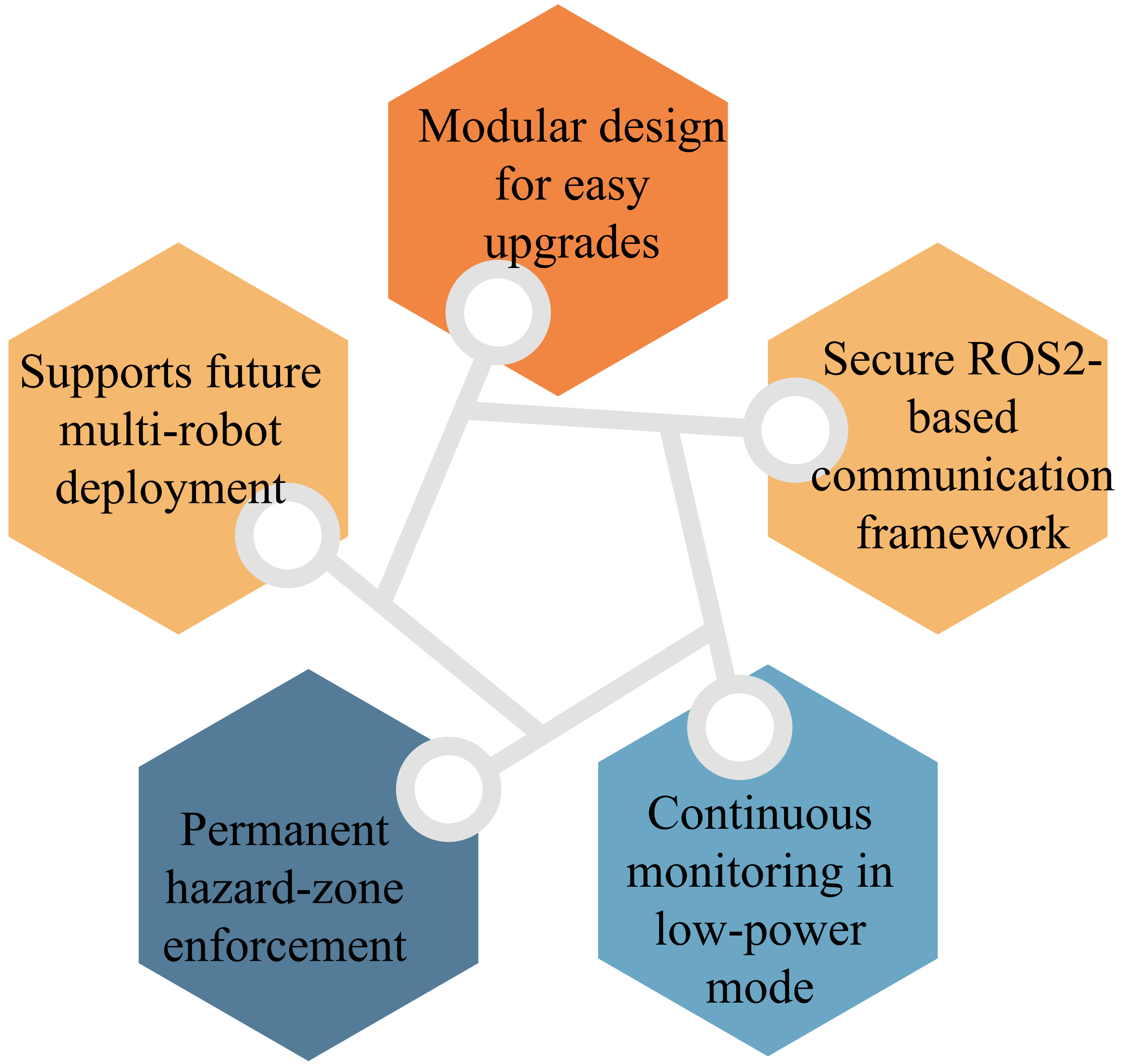
TECHNOLOGIES, TOOLS & FRAMEWORKS USED



METHODOLOGY

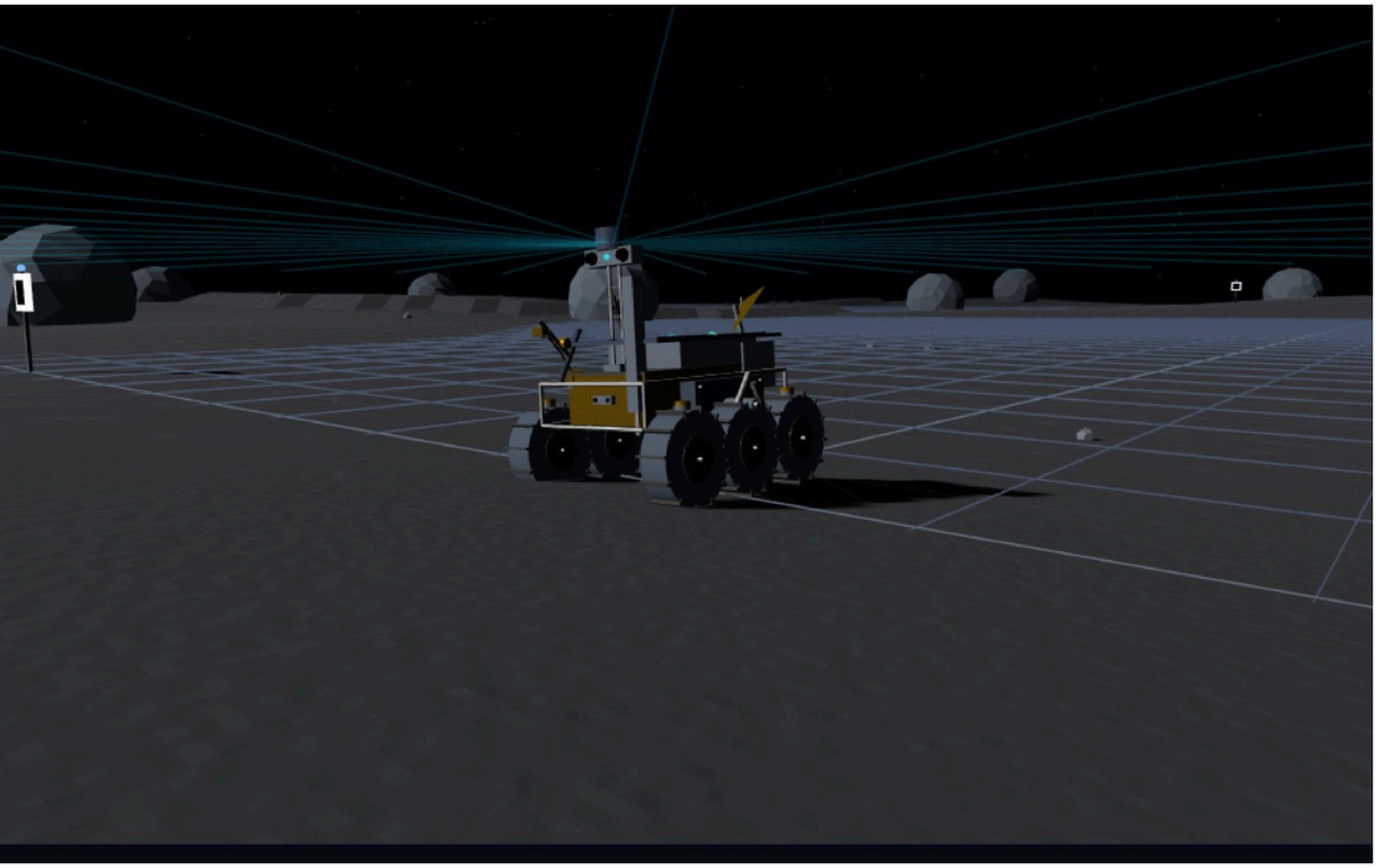
- Multi-sensor data collection using LiDAR, cameras, IMU, and environmental sensors
- Dust-filtered perception and SLAM-based mapping for accurate localization
- AI-driven anomaly detection for real-time hazard identification
- Autonomous navigation with dynamic hazard avoidance and energy-aware routing
- Explainable AI for transparent decision-making and astronaut interaction

SCALABILITY & SECURITY CONSIDERATIONS

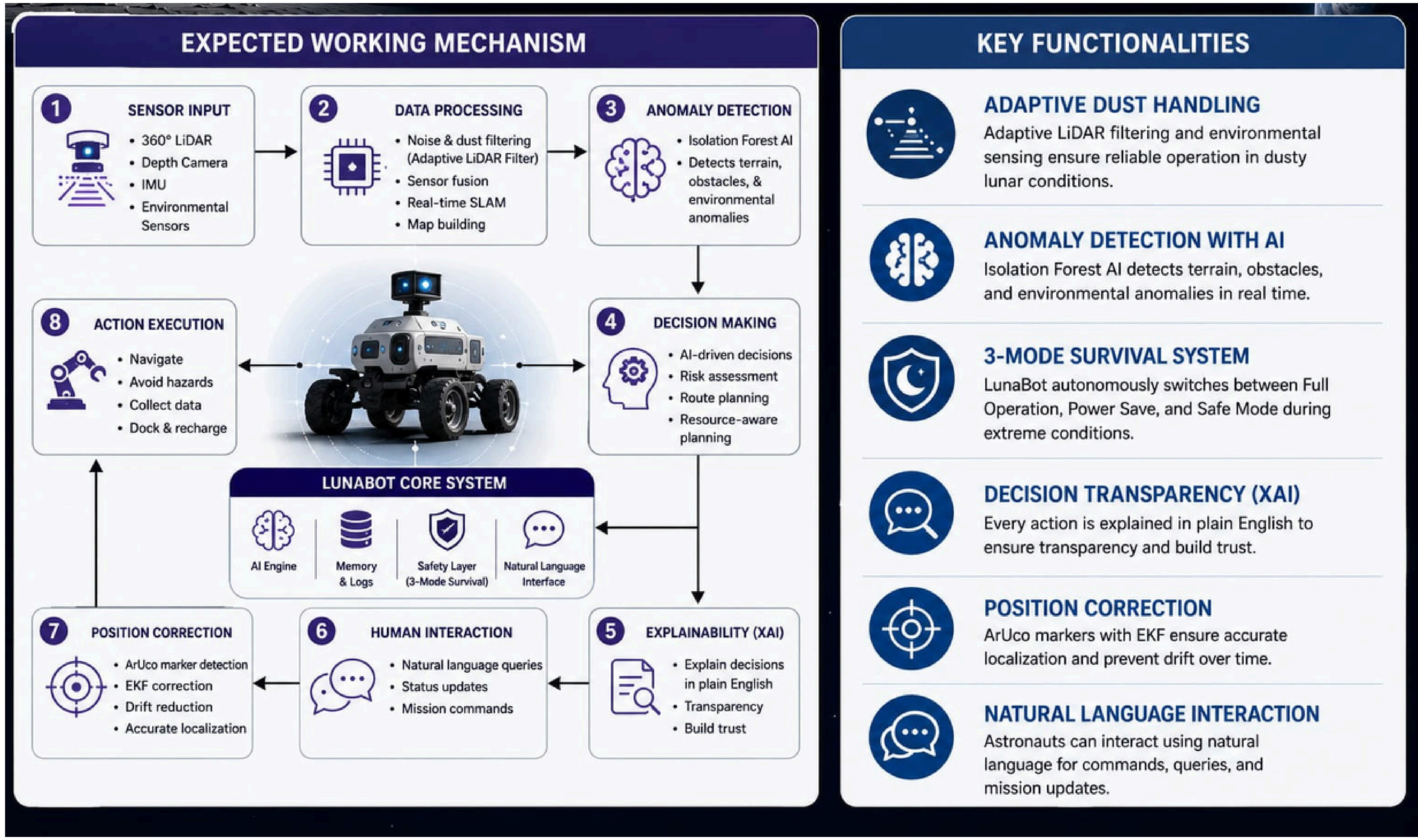


IMPLEMENTATION / PROTOTYPE

PROTOTYPE SCREENSHOT



EXPECTED WORKING MECHANISM & KEY FUNCTIONALITIES



DATASET DETAILS

SOURCE:

- NASA LADEE LDEX (lunar dust, 528MB, PDS)
- ·NASA LADEE NMS (lunar atmospheric gas composition)
- UCI Gas Sensor Array Drift Dataset (13,910 real multi-sensor readings, CC BY 4.0)
- UCI Gas Sensor Dynamic Mixtures (4.1M records, CO/CO_2/Ethylene)
- ·ROS2 LiDAR+IMU bag files (IEEE DataPort, 53,000+ messages)

DATASET USAGE

- UCI gas datasets train the Isolation Forest ML model on real chemical sensor anomalies.
- NASA LADEE thresholds define lunar-specific danger levels. ROS2 bag files validate SLAM and EKF localization pipeline.

FEASIBILITY & IMPACT

NEW HORIZON COLLEGE OF ENGINEERING

Technical Feasibility
- Built on ROS2 (Nav2, slam_toolbox, robot_localization)
- No custom complex algorithms required
- ML model trains < 10 sec, inference 1 ms
- System runs on a single laptop
- One launch command execution

Economic Viability
- Open-source software (no licensing cost)
- LLM API cost: very low (cents pr qry)
- Hardware prototype cost: ₹22,260
- Uses standard, easily available components

Operational Feasibility
- Fully autonomous system
- Single launch file startup
- No technical expertise required for usage
- Natural language interaction for astronauts

Scalability Feasibility
- Modular 5-layer architecture
- Works in simulation and real hardware
- Supports multi-robot deployment
- ML model easily retrainable

Safety Feasibility
- Monitoring always active (never stops)
- Permanent hazard zones enforced
- No single point of failure
- Full decision logging with explanations

EXPECTED SOCIAL / BUSINESS / ENVIRONMENTAL IMPACT

LunaBot delivers measurable impact across society, industry, and the lunar environment.

SOCIAL IMPACT
- **Enhances Astronaut Safety** – 24/7 autonomous monitoring and early hazard detection reduce mission risks.
- **Supports Human Exploration** – Enables long-duration missions by ensuring habitat and terrain safety.
- **Advances Knowledge & Education** – Generates lunar hazard maps and data that benefit global research and inspire future generations.
- **Promotes Global Collaboration** – Shares data and insights to support international partnerships in space exploration.
- **Inclusive & Accessible Interaction** – Natural language interface makes complex information easy to access for everyone.

BUSINESS IMPACT
- **Reduces Operational Costs** – Autonomous operation minimizes human intervention and mission downtime.
- **Increases Mission Efficiency** – Real-time hazard detection and routing optimize time, energy, and resources.
- **De-Risks Lunar Investments** – Reliable autonomous systems lower mission risk and increase return on investment.
- **Creates Market Opportunities** – Paves the way for lunar infrastructure services, data products, and robotic solutions.
- **Scalable & Future-Ready** – Modular architecture supports multi-robot deployments and future lunar missions.

ENVIRONMENTAL IMPACT
- **Minimizes Lunar Disturbance** – Autonomous navigation and precision mapping avoid unnecessary terrain disruption.
- **Monitors Lunar Environment** – Tracks dust, temperature, and other factors to protect habitats and equipment.
- **Energy Efficient Operations** – Smart energy-aware routing and night survival system reduce power consumption.
- **Sustainable Exploration** – Supports long-term human presence through safe, efficient, and responsible operations.
- **Enables Planetary Stewardship** – Builds technology and practices for preserving celestial environments.

REAL-WORLD APPLICABILITY
- Useful in space habitats, mines, nuclear plants, and disaster zones
- Enables safe operation in hazardous, GPS-denied environments

SCALABILITY & SUSTAINABILITY
- Modular architecture supports easy upgrades and multi-robot deployment
- Energy-efficient operation with low-power survival modes

EXPECTED IMPACT
- Improves safety for astronauts and workers in hazardous industries
- Reduces operational risk and monitoring workload through AI automation
- Supports development of sustainable lunar habitats and future space missions
- Enhances disaster management and emergency-response capabilities

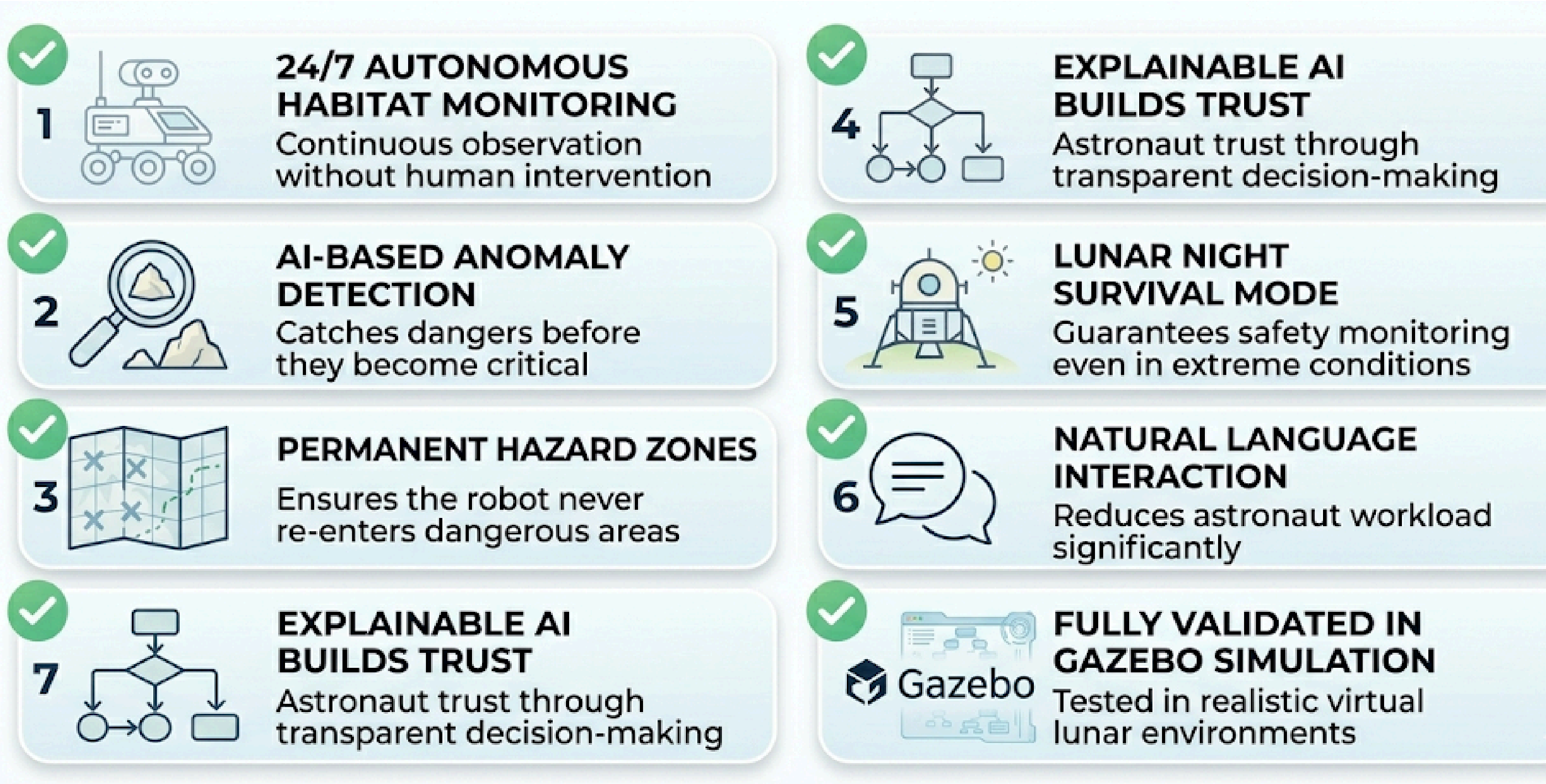
FUTURE ENHANCEMENTS
- Physical hardware deployment with onboard edge computing
- Multi-robot collaboration and swarm coordination
- Advanced predictive AI for early hazard forecasting
- Integration with real mission telemetry and space communication systems

CONCLUSION

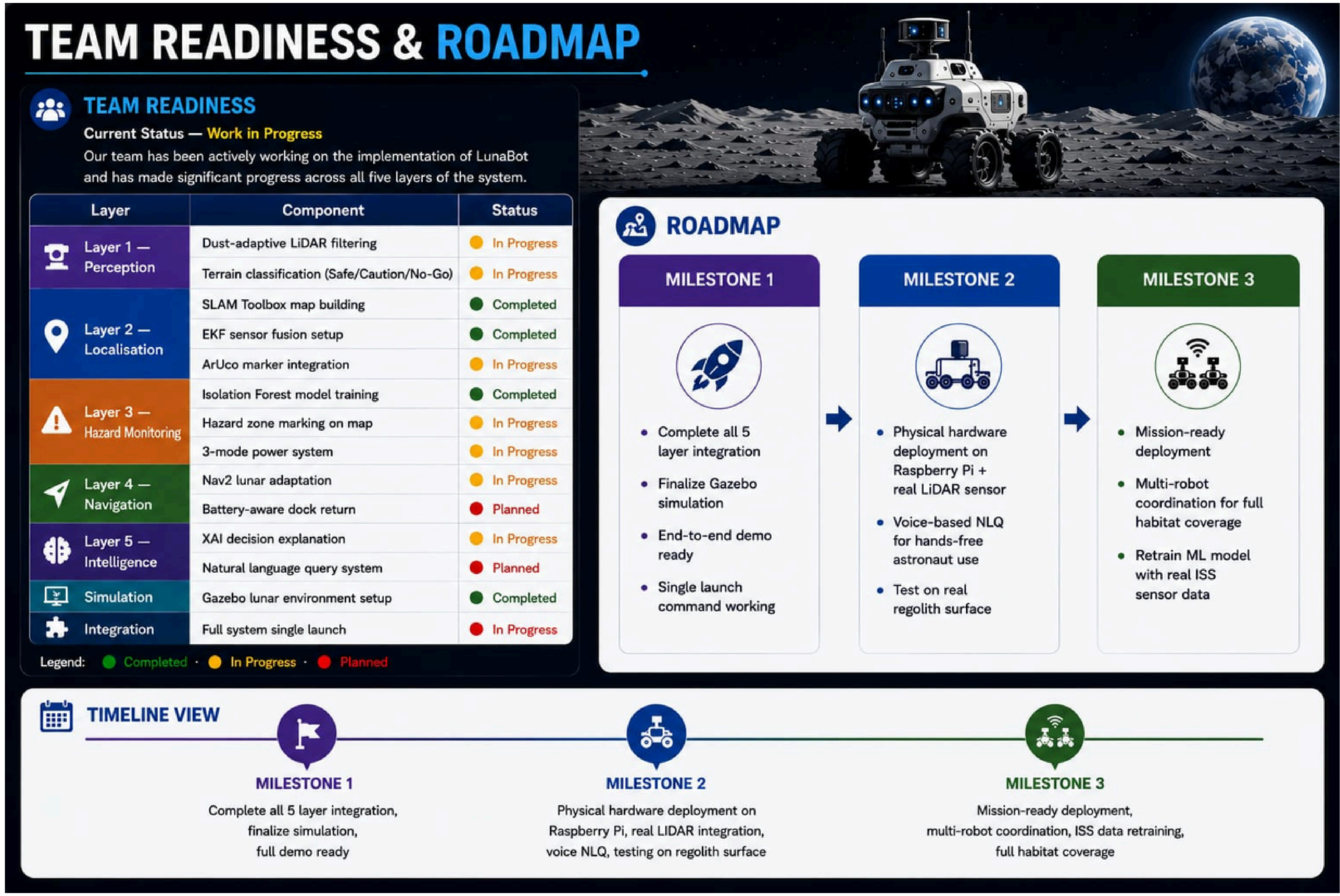
SUMMARY

- Autonomous AI-powered robot for safe lunar habitat monitoring and navigation
- Continuously tracks oxygen, CO_2, pressure, and temperature in real time
- Uses AI-based anomaly detection to identify hazards and create safety zones
- Performs intelligent, energy-efficient autonomous navigation with explainable decisions
- Ensures continuous astronaut safety and reduces dependence on human supervision

KEY OUTCOMMES & BENEFITS



TEAM READINESS AND ROADMAP



Layer	Component	Status
Layer 1 — Perception	Dust-adaptive LiDAR filtering	In Progress
	Terrain classification (Safe/Caution/No-Go)	In Progress
	SLAM Toolbox map building	Completed
Layer 2 — Localisation	EKF sensor fusion setup	Completed
	ArUco marker integration	In Progress
	Isolation Forest model training	Completed
Layer 3 — Hazard Monitoring	Hazard zone marking on map	In Progress
	3-mode power system	In Progress
Layer 4 — Navigation	Nav2 lunar adaptation	In Progress
	Battery-aware dock return	Planned
Layer 5 — Intelligence	XAI decision explanation	In Progress
	Natural language query system	Planned
Simulation	Gazebo lunar environment setup	Completed
Integration	Full system single launch	In Progress

Thank You!

"Vigilence that never sleeps even on moon!"